

SHOPNEST SALES ANALYTICS DASHBOARD

Power BI Capstone Project Report

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PROJECT OVERVIEW

The ShopNest Sales Analytics Dashboard was developed using Power BI to analyze and visualize key business metrics from the ShopNest e-commerce dataset. The dashboard provides insights into sales performance, revenue trends, product ratings, payment methods, delivery efficiency, and customer purchasing behavior.

Through interactive visualizations and slicers, users can explore data by Customer State, Year, and Product Category, enabling dynamic and detailed analysis. The project aims to support data-driven decision-making by identifying sales trends, operational challenges, and growth opportunities across different business areas.

The dashboard delivers a clear and user-friendly view of business performance, helping stakeholders monitor key metrics and make informed strategic decisions.

Tools Used:

- Power BI Desktop
- Data Modeling
- DAX Measures
- Data Visualization

Dashboard Features

- Interactive and user-friendly dashboard design.
- Dynamic filtering using slicers:
- Customer State

- Year
- Product Category Name
- Analysis of Top Product Categories by Sales.
- State-wise Sales Performance Analysis.
- Delayed vs On-Time Order Tracking.
- Monthly and Quarterly Sales Trend Analysis.
- Payment Method Distribution Analysis.
- Product Rating Analysis (Top & Bottom Rated Products).
- Revenue Trend Analysis.
- Drillthrough functionality for detailed order analysis.
- Responsive visualizations for easy business performance monitoring.
- Data-driven insights to support strategic decision-making.

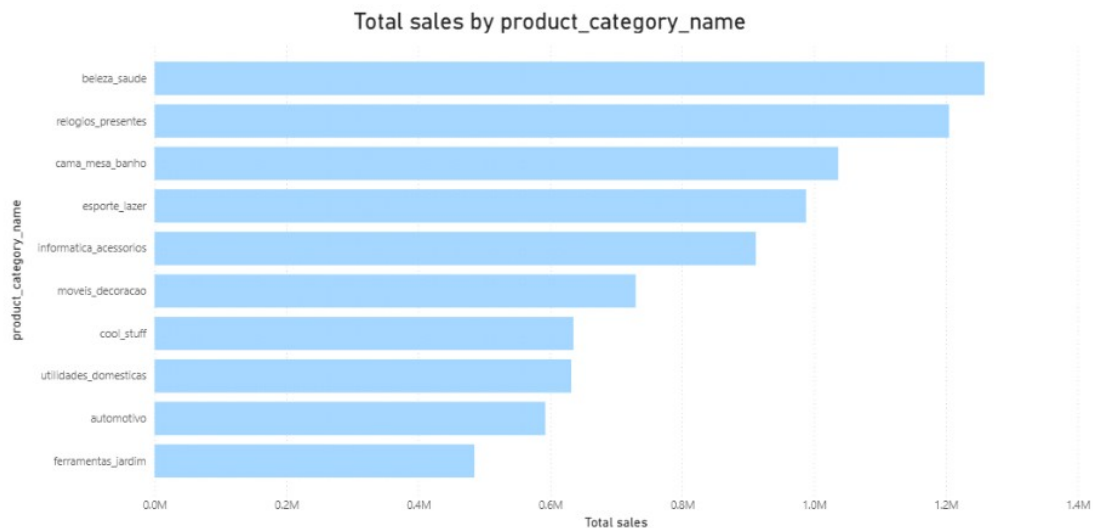
The dashboard enables users to explore business performance from multiple perspectives and gain actionable insights through interactive visual analytics.

QUESTION 1

Top Categories by Total Price:

Identify and visually represent the top 10 product categories by total sales.

Visualization: Bar Chart showing the Top 10 Product Categories ranked by total sale amount.



Explanation:

This visualization highlights the ten highest-performing product categories based on their total sales value. The categories are ranked in descending order, allowing easy identification of the products contributing the most revenue to the business.

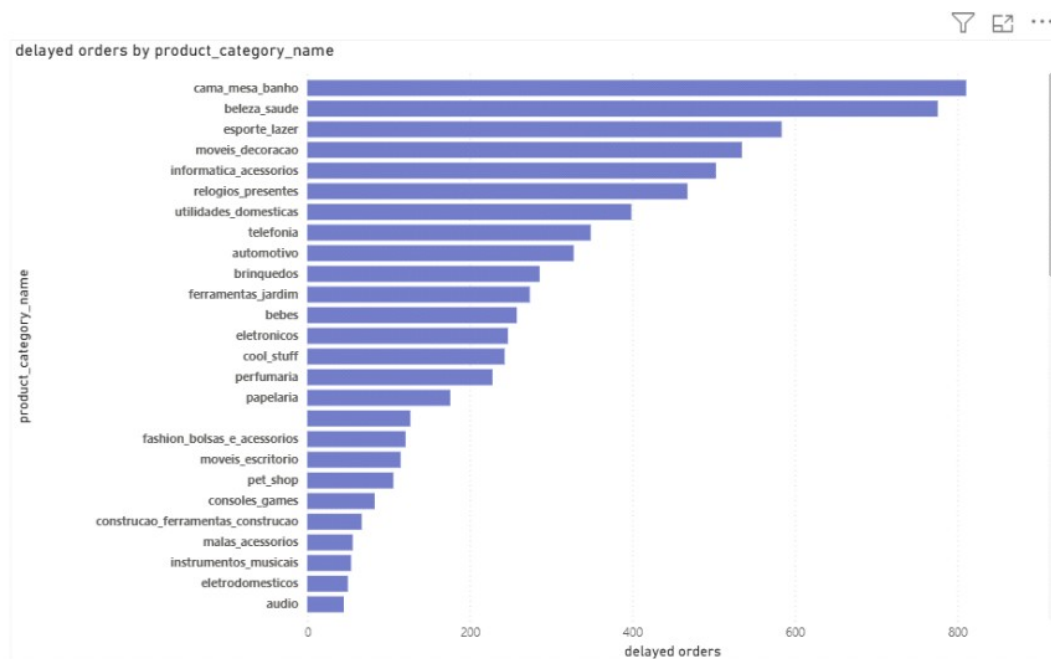
The analysis reveals that the leading categories generate a significant share of overall sales, indicating strong customer demand and market performance. By comparing category-wise sales, stakeholders can quickly identify high-revenue segments and prioritize inventory planning, marketing campaigns, and business strategies accordingly.

QUESTION 2

Delayed Orders Analysis:

Determine the number of delayed orders in each category. An order is considered delayed if the actual delivery date is later than the estimated delivery date.

Visualization: Bar Chart displaying the count of delayed orders across different product categories.



Ex

planation:

This visualization analyzes delivery performance by identifying orders that were delivered after their estimated delivery date. The delayed orders are grouped by product category to understand which categories experience the highest delivery delays.

By examining the distribution of delayed orders, the business can identify operational challenges, supply chain bottlenecks, or logistics issues affecting specific product categories. This insight helps improve delivery efficiency and customer satisfaction by enabling targeted corrective actions.

QUESTION 3

Monthly Comparison of Delayed and On-Time Orders:

Create a dynamic visual that compares the number of delayed orders to the number of on-time orders for each month.

Utilize the drillthrough feature for detailed analysis of late and on-time deliveries.

Visualization: Clustered Column Chart comparing Delayed Orders and On-Time Orders on a monthly basis, with Drillthrough functionality enabled for detailed order analysis.



Explanation:

This visualization provides a month-by-month comparison of orders delivered on time versus those delivered after the estimated delivery date. It helps track delivery performance trends over time and identify periods where delays were more frequent.

The dynamic nature of the visual allows users to interact with the data and observe changes in delivery performance across different months. Additionally, the drillthrough feature enables deeper investigation into specific months, allowing users to view detailed information about delayed and on-time deliveries, including order-level insights.

Key Insights:

Compares delayed and on-time order counts for each month.

Highlights seasonal trends or periods with increased delivery delays.

Helps evaluate overall logistics and fulfillment performance over time.

Drillthrough functionality provides detailed order-level analysis for further investigation.

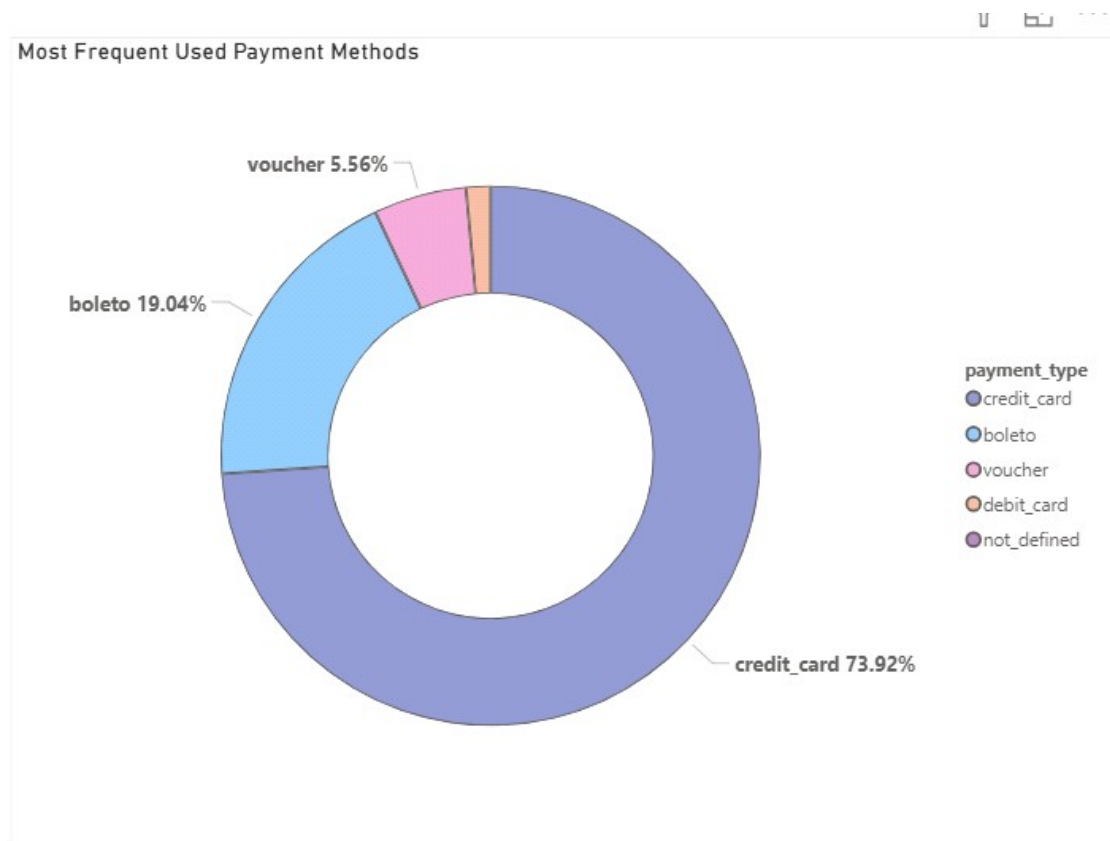
Enables stakeholders to make informed decisions to enhance customer satisfaction and operational efficiency.

QUESTION 4

Payment Method Analysis:

Analyze the most frequently used payment methods by customers using a visually appealing representation, such as a pie chart or other suitable visual.

Visualization: Donut Chart displaying the distribution of customer transactions across different payment methods.



Ex

planation:

This visualization illustrates the proportion of orders completed using each payment method through a donut chart. The chart provides a clear and visually engaging representation of customer payment preferences, making it easy to identify the most frequently used payment options.

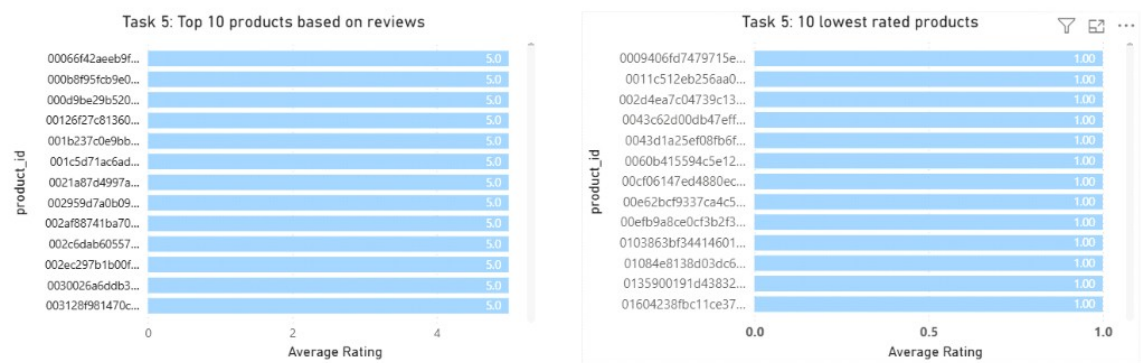
The donut chart highlights the contribution of each payment method to the total number of transactions, enabling stakeholders to understand customer purchasing behavior and payment trends. This analysis helps businesses optimize payment services and enhance the overall checkout experience.

QUESTION 5

Product Rating Analysis:

Determine the top 10 highest-rated products and the bottom 10 lowest-rated products using a bar or column chart.

Visualization: Bar Chart displaying the Top 10 Highest-Rated Products and Bottom 10 Lowest-Rated Products based on customer ratings.



Ex

planation:

This visualization evaluates product performance through customer ratings by identifying the ten highest-rated products and the ten lowest-rated products. The bar chart provides a clear comparison of product ratings, making it easy to distinguish products that are highly appreciated by customers from those receiving lower satisfaction scores.

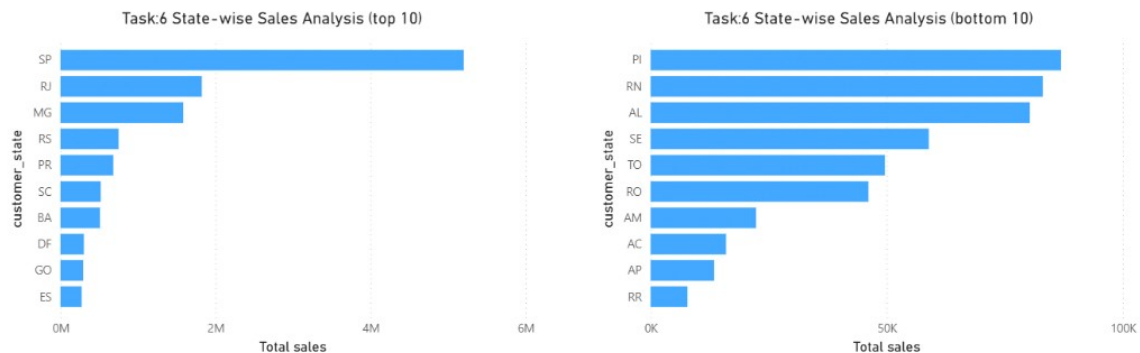
The analysis helps businesses understand customer preferences, identify successful products, and recognize products that may require quality improvements or further investigation. By comparing both ends of the rating spectrum, stakeholders can gain valuable insights into overall product performance and customer satisfaction levels.

The chart enables quick identification of products that consistently receive positive feedback as well as products that may be negatively impacting the customer experience, supporting data-driven decisions for product development, inventory management, and quality enhancement initiatives.

Question 6

Identify and visually represent states with high and low sales, providing a clear understanding of regional sales performance.

Visualization: Bar Chart displaying total sales across different states.



Ex

planation:

This visualization presents the distribution of sales across various states, enabling a comprehensive analysis of regional business performance. The bar chart compares total sales generated by each state, making it easy to identify regions that contribute the highest and lowest revenue.

By analyzing state-wise sales trends, businesses can understand geographical demand patterns, evaluate market performance, and identify opportunities for expansion in underperforming regions. The visualization helps stakeholders quickly compare sales figures across states and assess the effectiveness of regional sales strategies.

The analysis supports data-driven decision-making by highlighting strong-performing markets as well as areas that may require additional marketing efforts, customer engagement initiatives, or operational improvements to increase sales performance.

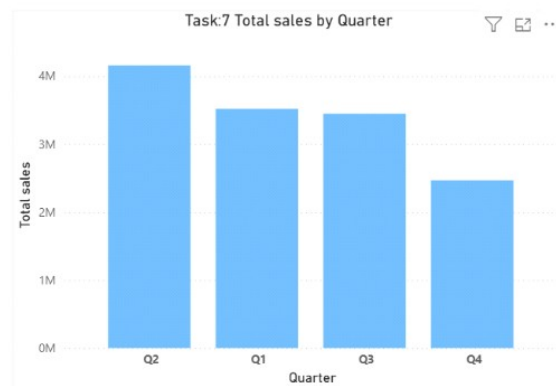
Question 7

Investigate and visualize any seasonal patterns (quarterly) or trends in sales data over the course of the year.

Visualization:

Line Chart showing Month-wise Sales Trend.

Clustered Column Chart displaying Total Sales by Quarter.



Ex

planation:

This analysis examines sales performance over time to identify seasonal trends and fluctuations throughout the year. A line chart is used to visualize month-wise sales trends, providing a detailed view of how sales change from one month to another. This helps identify periods of growth, decline, or consistent performance.

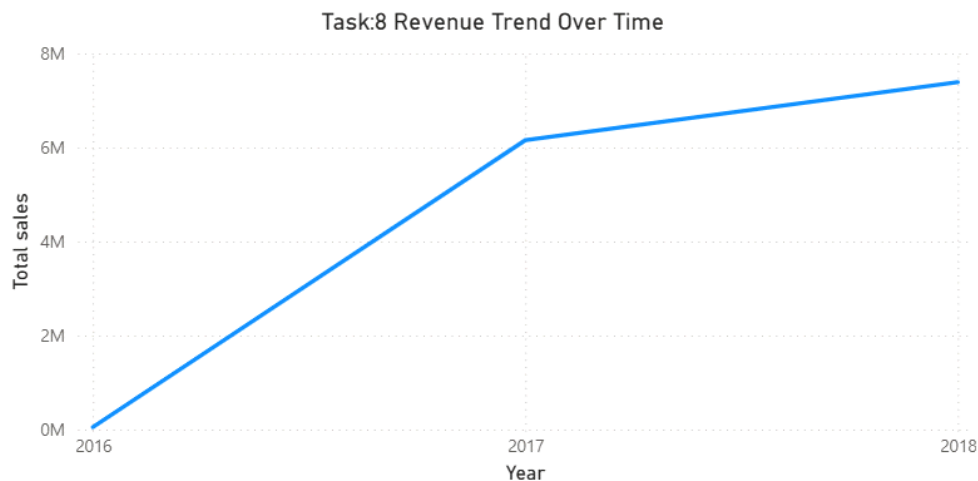
To further understand seasonal behavior, a clustered column chart is used to compare total sales across quarters. This visualization highlights which quarters generated the highest and lowest sales, making it easier to identify broader seasonal patterns and business cycles.

Together, these visuals provide a comprehensive view of sales performance by combining detailed monthly trends with high-level quarterly comparisons. This analysis helps stakeholders understand customer purchasing patterns, forecast future demand, optimize inventory planning, and make informed business decisions based on seasonal sales behavior.

Question 8

Determine the total revenue generated by ShopNest Store and analyze how it changes over time (Yearly). Represent this information through suitable visuals to highlight trends and patterns.

Visualization: Line Chart showing Year-wise Revenue Trend.



Ex

planation:

This analysis examines the revenue generated by ShopNest Store and how it changes over time. A line chart is used to visualize yearly revenue trends, providing a clear view of the store's financial performance across different years.

The visualization helps identify periods of revenue growth, decline, or stability, enabling stakeholders to understand the overall business trajectory. By tracking revenue over time, it becomes easier to evaluate business performance, recognize long-term trends, and assess the effectiveness of sales strategies.

The year-wise comparison allows for a better understanding of revenue patterns and supports data-driven decision-making related to forecasting, business planning, and future growth opportunities. This analysis provides valuable insights into the store's financial progress and overall sales performance.

DASHBOARD INTERACTIVITY

To enhance the user experience and enable dynamic data exploration, interactive slicers have been incorporated into the dashboard. These slicers allow users to filter and analyze the data based on specific criteria, making the dashboard more flexible and insightful.

Slicers Used

1. Customer State

Allows users to filter the dashboard based on customer location.

Helps analyze sales performance, order trends, delivery performance, and customer behavior for specific states.

2. Year

Enables filtering of data for a particular year.

Supports year-wise comparison of sales, revenue, orders, and other key business metrics.

3. Product Category Name

Allows users to focus on specific product categories.

Helps evaluate category-wise sales, ratings, payment preferences, and delivery performance.

Interactivity Benefits

All visuals update dynamically based on the selected slicer values.

Users can perform focused analysis without modifying the underlying data.

Enhances dashboard usability and supports deeper business insights through customized filtering.

Provides a more interactive and user-friendly reporting experience, enabling efficient exploration of sales and operational performance across multiple dimensions.

CONCLUSION

The ShopNest Sales Analytics Dashboard provides a comprehensive overview of the company's sales performance, customer behavior, delivery efficiency, product ratings, payment preferences, and revenue trends. Through a combination of interactive visualizations and dynamic filters, the dashboard transforms raw transactional data into meaningful business insights.

The analysis identified top-performing product categories, regional sales performance across states, seasonal sales trends, customer payment preferences, and product rating patterns. Additionally, delivery performance was evaluated by comparing delayed and on-time orders, helping to highlight operational strengths and areas for improvement.

The implementation of interactive slicers for Customer State, Year, and Product Category enables users to explore the data from multiple perspectives and perform detailed analysis based on their specific requirements.

Overall, the dashboard serves as an effective decision-support tool that helps stakeholders monitor business performance, identify growth opportunities, optimize operations, and make data-driven decisions to improve customer satisfaction and drive future business success..